Development of a computational strategy to search for pharmacochaperones of the GALT p.Q188R enzyme

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INTRODUCTION

Classic galactosemia is a rare inborn error of metabolism associated with mutations that affect the activity and the stability of GALT. To date, there is no pharmacological treatment for this disease. Recently, we developed a computational approach to identify candidate pharmacochaperones (PCs) for GALT p.Q188R (the most common mutant enzyme in classic galactosemic patients).

COMPUTATIONAL STRATEGY

Selection of 5 putative pharmacochaperones

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Molecular Docking on Central cavity of wild-type and p.Q188R GALT

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1. Study in literature;
2. MD simulations;
3. Use of a predictor software

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Identification of two putative allosteric sites in GALT

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Molecular Docking of the 5 PCs on the two predicted allosteric sites of wild-type and p.Q188R GALT

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Extraction of pharmacophore models and research of pharmacological hits in DrugBank database

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Identification of additional 19 putative PCs

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Molecular Docking of the 19 PCs on the two predicted allosteric sites of wild-type and p.Q188R GALT

RESULTS

Thanks to this approach, we selected six promising candidate PCs (Fig. 1) using molecular docking studies on two possible allosteric sites, pharmacophore modeling and virtual screening.

In vitro studies are ongoing to test the binding of these molecules to GALT enzyme, with promising preliminary results.

Binding Energy kcal/mol	GALT	PC1	PC2	PC5	PC7	PC10	PC11
	WT	-7.62	-8.36	-9.49	-10.89	-8.05	-9.59
	p.Q188R	-7.47	-8.2	-9.54	-10.97	-8.26	-9.00

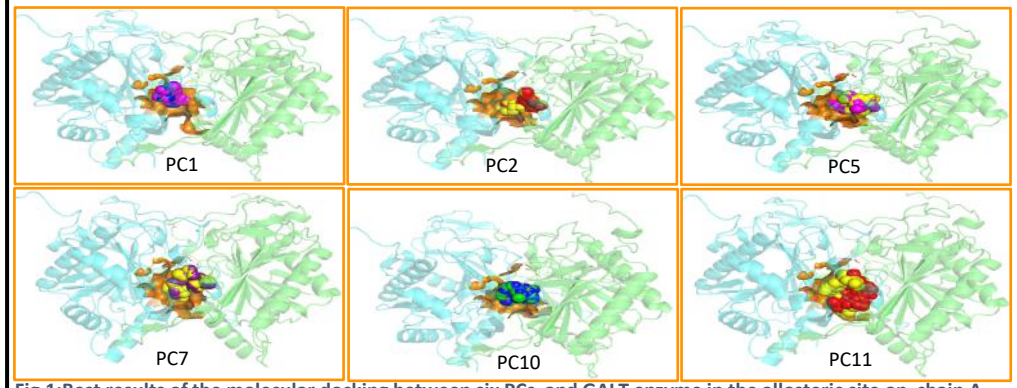
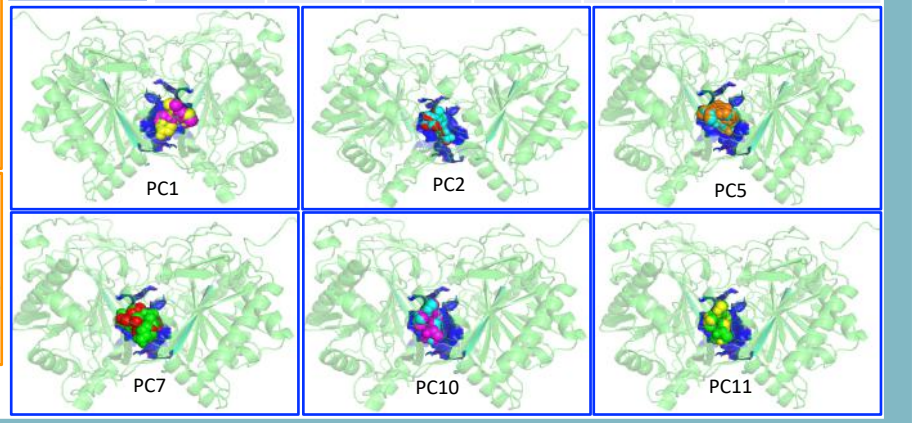


Fig.1:Best results of the molecular docking between six PCs and GALT enzyme in the allosteric site on chain A (orange) and on chain B (blu)

Binding Energy kcal/mol	GALT	PC1	PC2	PC5	PC7	PC10	PC11
	WT	-7.26	-7.24	-7.64	-10.49	-7.66	-7.95
	p.Q188R	-7.03	-7.56	-9.54	-10.54	-7.8	-6.9